30 The following process shows a stationary isotope of boron, which upon capturing a slow-moving neutron, splits to form a lithium isotope and an alpha particle.

$$^{10}_{5}\mathrm{B} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{7}_{3}\mathrm{Li} + {}^{4}_{2}\mathrm{He} + \gamma$$

The nuclear binding energies per nucleon of the isotopes are:

$^{10}_{5}\mathrm{B}$: 6.494 MeV
$^{7}_{3}\mathrm{Li}$: 5.607 MeV
$^{4}_{2}\mathrm{He}$: 7.120 MeV

What is the energy of the γ-ray emitted, given that the total kinetic energy of $^{7}_{3}\mathrm{Li}$ and $^{4}_{2}\mathrm{He}$ is 2.31 MeV?

A 0.48 MeV **B** 2.79 MeV **C** 5.10 MeV **D** 6.23 MeV